HHCKLA Buddhist Leung Chik Wai College

6A error analysis (2024)

Student No.: 18 Name: _____ Marks: / 55

Class: 6A Date: _____ Marked date: _____

What have you learned in this chapter? _____

What can you do to improve your work next time? _____

1. The base of a solid right pyramid is a square of side 64 cm. The height of the pyramid is 24 cm. The pyramid is divided into a frustum X and a pyramid Y by a plane which is parallel to its base. It is given that the height of Y is 18 cm.

   (a) Find the volume of X. (3 marks)

   (b) The base of another solid right pyramid is a square. This pyramid is divided into a frustum Z and a pyramid by a plane which is parallel to its base. The height and the total surface area of Z are 3 cm and 960 cm^2 respectively. Are X and Z similar? Explain your answer. (4 marks)

   (7 marks) (DSE-2024 PI Q13)

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2. Let $F(x) = (6x^2 + x + p)(qx^2 + rx - 10)$, where p, q and r are constants. The constant term of $F(x)$ is 40.

(a) Write down the value of p. (1 mark)

(b) When $F(x)$ is divided by $x + 1$, the remainder is -12. It is given that $x - 2$ is a factor of $F(x)$.

(i) Find q and r.

(ii) How many irrational roots does the equation $F(x) = 0$ have? Explain your answer. (7 marks)

(8 marks) (DSE-2024 PI Q14)

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3. (a) $PQRS$ is a thin quadrilateral metal sheet, where $PQ = 12\,\text{cm}$, $PS = 10\,\text{cm}$, $QR = 13\,\text{cm}$, $\angle QPS = 82^\circ$ and $\angle QRS = 65^\circ$. Find

(i) the length of QS,

(ii) $\angle RQS$. (4 marks)

(b) The metal sheet $PQRS$ described in (a) is now folded along QS (see Figure 2). It is given that the angle between the plane PQS and the plane QRS is 80°.

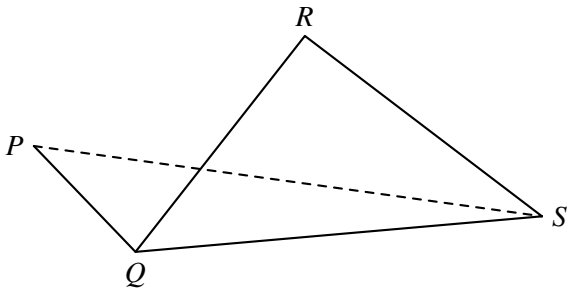


Figure 2

(i) Find the shortest distance from R to the plane PQS.

(ii) Let X be any point lying on the plane QRS. Someone claims that the distance between P and X exceeds $8\,\text{cm}$. Is the claim correct? Explain your answer. (4 marks)

(8 marks) (DSE-2024 PI Q18)

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4. Let $f(x) = 2x^2 + 4mx + 8x + 2m^2 + 8m + n$, where m and n are real constants such that $mn < 0$. Denote the vertex of the graph of $y = f(x)$ by P.

(a) Using the method of completing the square, express the coordinates of P in terms of m and n. (2 marks)

(b) Describe the geometric meaning represented by transforming $f(x)$ to $f\left(\frac{x}{5}\right) + 7$. (2 marks)

(c) Denote the vertex of the graph of $y = f\left(\frac{x}{5}\right) + 7$ by Q. Let (a_1, b_1) and (a_2, b_2) be the coordinates of P and Q respectively. It is given that a_1, $1 + n$, a_2 is an arithmetic sequence and b_1, $4 - m$, b_2 is a geometric sequence.

(i) Find the coordinates of P and Q.

(ii) The coordinates of the points R and S are $(3t + 27, t)$ and $(3t + 3, 2t - 3)$ respectively, where t is a real number. Is it possible that $PQRS$ is a rhombus? Explain your answer. (8 marks)

(12 marks) (DSE-2024 PI Q19)

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5. If $k = \dfrac{5}{2m} + n$, then $m =$

A. $\dfrac{5}{2(k-n)}$.

B. $\dfrac{5}{2(n-k)}$.

C. $\dfrac{2(k-n)}{5}$.

D. $\dfrac{2(n-k)}{5}$.

(2 marks) (DSE-2024 PII Q03)

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6. In a school, 40% of the students are girls and $\beta\%$ of the girls are foreign students. It is given that 30% of the boys in the school are foreign students. In the school, the number of foreign students and the number of girls are equal. Find β.

A. 20

B. 45

C. 55

D. 80

(2 marks) (DSE-2024 PII Q11)

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7. Which of the following statements about the graph of $y = 2(6-x)^2 - 7$ is true?

A. The graph opens upwards.

B. The graph does not cut the x-axis.

C. The y-intercept of the graph is -7.

D. The graph passes through the point $(-6, -7)$.

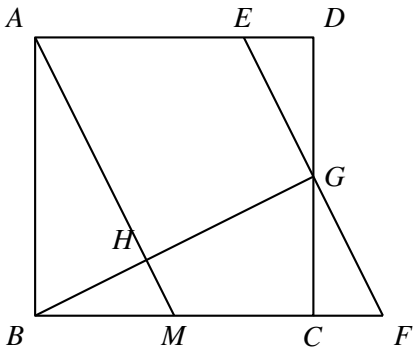
(2 marks) (DSE-2024 PII Q14)

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8. In the figure, $ABCD$ is a square. Let M be the mid-point of BC. E is a point lying on AD such that $AE : ED = 3 : 1$. F is a point lying on BC produced such that $EF//AM$. CD and EF intersect at the point G while AM and BG intersect at the point H. If the area of $\triangle BMH$ is $4\,\text{cm}^2$, then the area of the trapezium $AEGH$ is



A. $12\,\text{cm}^2$.

B. $33\,\text{cm}^2$.

C. $39\,\text{cm}^2$.

D. $45\,\text{cm}^2$.

(2 marks) (DSE-2024 PII Q17)

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9. If the sum of the interior angles of a regular polygon is 900°, which of the following is/are true?

I. The number of diagonals of the polygon is 7.

II. The number of folds of rotational symmetry of the polygon is 7.

III. The number of axes of reflectional symmetry of the polygon is 7.

A. I only

B. II only

C. I and III only

D. II and III only

(2 marks) (DSE-2024 PII Q20)

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10. $AF000000000BC_{16} =$

A. $175 \times 16^{11} + 188$.

B. $192 \times 16^{11} + 205$.

C. $175 \times 16^{12} + 188$.

D. $192 \times 16^{12} + 205$.

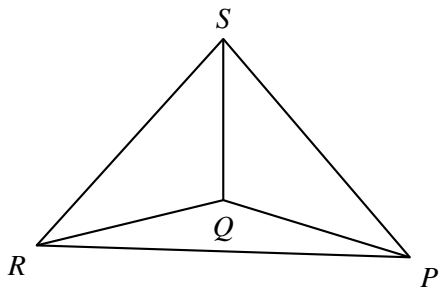
(2 marks) (DSE-2024 PII Q32)

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11. In the figure, the base PQR of the tetrahedron $PQRS$ lies on the horizontal ground. It is given that Q is vertically below S. If $\angle PQR = 90°$, $\angle QPS = 30°$ and $\angle QRS = 45°$, then $\cos\angle PRS =$



A. $\frac{1}{2}$.

B. $\frac{3}{4}$.

C. $\frac{\sqrt{2}}{4}$.

D. $\frac{\sqrt{3}}{6}$.

(2 marks) (DSE-2024 PII Q40)

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12. Let G, H, I and J be the centroid, the orthocentre, the in-centre and the circumcentre of $\triangle PQR$ respectively. If $\angle PQR = \angle PRQ = 22°$, which of the following are true?

I. G lies inside $\triangle PQR$.

II. H lies outside $\triangle PQR$.

III. I, J and Q are collinear.

A. I and II only

B. I and III only

C. II and III only

D. I, II and III

(2 marks) (DSE-2024 PII Q41)

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13. There are three questions in a test. The probabilities that a student answers the first question correctly, the second question correctly and the third question correctly are 0.6, 0.7 and 0.8 respectively. Find the probability that the student answers at least 1 question correctly in this test.

A. 0.024

B. 0.188

C. 0.812

D. 0.976

(2 marks) (DSE-2024 PII Q43)

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14. If the variance of the seven numbers x_1, x_2, x_3, x_4, x_5, x_6 and x_7 is 16, then the standard deviation of the seven numbers $9x_1 - 5$, $9x_2 - 5$, $9x_3 - 5$, $9x_4 - 5$, $9x_5 - 5$, $9x_6 - 5$ and $9x_7 - 5$ is

A. 31.

B. 36.

C. 139.

D. 144.

(2 marks) (DSE-2024 PII Q45)

I can finish _____ questions

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